

Week 01

Task 01

Lambda function to resize image

For Least privileges policy

Resize Lambda:

- Can only read originals/*
- Can only write resized/*

Cleanup Lambda:

- Can only list the target bucket
- Can only delete resized/*
- Cannot delete the whole bucket

Challenges:

A lot of microservices lead to many steps which are hard to remember as a beginner.

The resize function used Pillow, hence we needed to package dependencies. AWS supports deploying Python Lambda functions using .zip packages

Version conflicts due to different pythons in local system and AWS.

START



Create S3 Buckets

(original-images-task1-2026)

(resized-images-task1-2026)



Create Custom IAM Policies

(LambdaImageResizePolicyTask1)

(LambdaCleanupBucketPolicy)



Create IAM Roles

(resize-lambda-role)

(cleanup-lambda-role)



Attach Policies to Roles



Create Resize Lambda Function

(resize-image-lambda)



Create S3 Trigger



TEST WORKFLOW

Upload image to originals/Happyface.jpg



S3 Trigger fires



resize-image-lambda runs



Resized image saved to resized-images-task1-2026 (Bucket)



Create Cleanup Lambda Function

(empty-resized-bucket-lambda)



Manual Test Cleanup Lambda



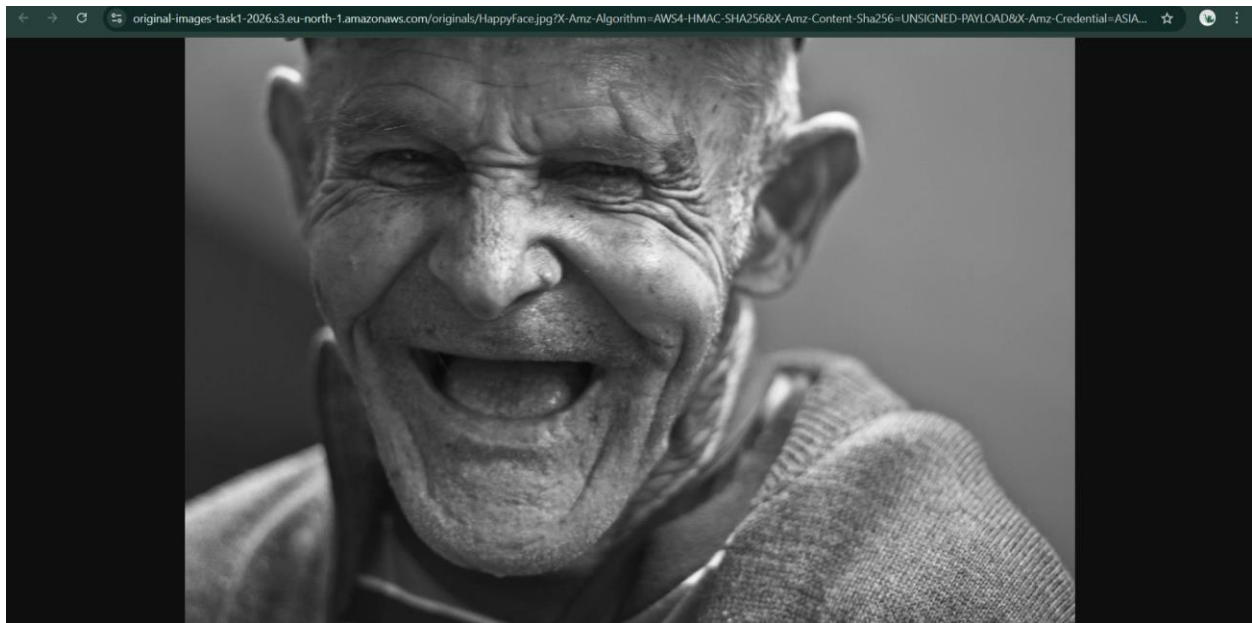
Bucket empties successfully



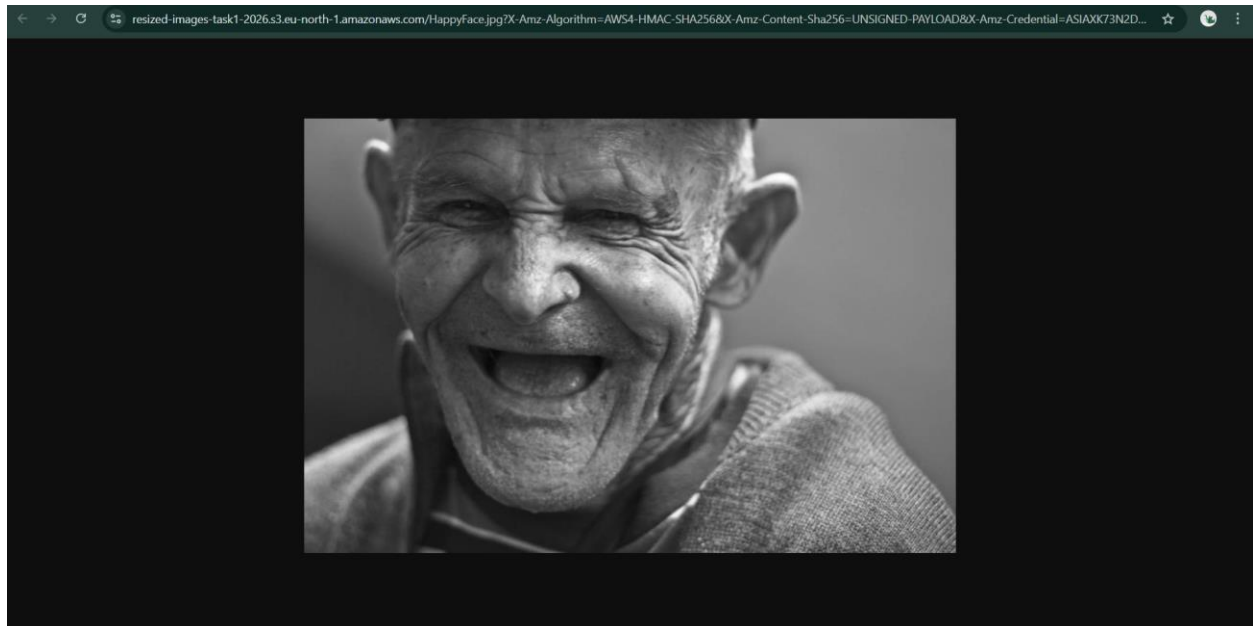
Create EOD Cron Job
temporary-empty-resized-bucket-eod
↓
EventBridge Scheduler
(Timezone: Asia/Karachi)
↓
Triggers Cleanup Lambda at EOD
↓
Resized Bucket Emptied
↓
Verify Success
↓
Delete Scheduler (One time only)

↓
END

Before:



After:



Before Cron job:

resized-images-task1-2026 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems - new](#) | [Access Points](#)

Objects (1) [Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

☐	Name	Type	Last modified	Size	Storage class
☐	HappyFace.jpg	jpg	April 29, 2026, 22:31:14 (UTC+05:00)	42.9 KB	Standard

After Cron job:

resized-images-task1-2026 [Info](#)

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - *new*

Access Points

Objects (0)



Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 > ⚙

	Name	Type	Last modified	Size	Storage class
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No objects

You don't have any objects in this bucket.

Upload

The EventBridge Scheduler was configured to delete itself after running the scheduled cleanup job.

Schedules (0)



Disable

Edit

Delete

Create schedule

Search loaded schedules

All states

All groups

< 1 > ⚙

	Schedule name	Schedule group	Status	Target	Target type	Last modified
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No schedules

No schedules associated with this resource.

Create schedules